Quaternion vector rotation vs reference frame transformation

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Report

Formulas for rotating a vector by and axis/angle using a quaternion, vs transforming a vector from one reference frame to another

Quaternion Vector Rotation vs. Frame Transform

Rotating a Vector (Active Transformation)

To rotate vector **v** by angle θ around unit axis **n̂**:

Form the quaternion:

$$q = \cos\frac{\theta}{2} + \sin\frac{\theta}{2}(n_x\mathbf{i} + n_y\mathbf{j} + n_z\mathbf{k})$$

Apply the rotation (sandwich product):

$$\mathbf{v}' = q\,\mathbf{v}\,q^{-1} = q\,\mathbf{v}\,q^*$$

Where **v** is treated as a pure quaternion $0 + v_x\mathbf{i} + v_y\mathbf{j} + v_z\mathbf{k}$, and since q is unit, $q^{-1} = q^*$
.

The result **v'** is the same vector **v** rotated by θ around **n̂**, expressed in the **same frame**.

Transforming a Vector Between Frames (Passive Transformation)

If frame B is oriented relative to frame A by the same quaternion q (i.e., B's axes are rotated by θ around **n̂** relative to A), then to express vector **v** (known in frame A) in frame B:

$$\mathbf{v}^B = q^*\,\mathbf{v}^A\,q$$

This is the **conjugate sandwich** — the inverse operation.

Side-by-Side Summary

Operation	Formula	What changes
Rotate vector by q	$q\mathbf{v}q^*$	The vector moves; frame stays
Express vector in rotated frame	$q^*\mathbf{v}q$	The frame moves; vector stays

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Intuition

These are **inverses of each other**, which makes sense:

- If you rotate a vector **+30°**, its components in the original frame change as if the frame rotated **−30°** around it.
- Rotating the frame by q is equivalent to seeing the vector as rotated by q^{-1}.

This is the **active vs. passive** transformation duality — the same quaternion q describes both, just applied in opposite order.

Practical Note (Conventions vary!)

Some libraries/textbooks define q as rotating **from B to A** (i.e., $q_{A\leftarrow B}$), in which case:

$$\mathbf{v}^A = q\,\mathbf{v}^B\,q^* \quad \Longleftrightarrow \quad \mathbf{v}^B = q^*\,\mathbf{v}^A\,q$$

Always check whether a given quaternion is defined as a **rotation** or a **frame orientation**, and in which direction — this is the most common source of sign/conjugate errors.

Can you generally explain active vs passive rotations?

Active vs. Passive Rotations

These are two fundamentally different interpretations of the same mathematical object (a rotation matrix or quaternion).

Active Rotation

The object moves; the coordinate frame stays fixed.

You have a vector **v** sitting in a frame, and you physically rotate it to a new orientation **v'**. The basis vectors $\hat{x}$, $\hat{y}$, $\hat{z}$ don't move.

> *"Rotate the arrow 30° counterclockwise."*

Passive Rotation

The coordinate frame rotates; the object stays fixed.

The same physical vector exists in space, but you're now describing it using a new, rotated frame. The vector hasn't moved — you've just relabeled it.

> *"Describe the same arrow using axes rotated 30° counterclockwise."*

The Key Relationship

A **passive** rotation by +θ gives the same new components as an **active** rotation by **−θ**. They are inverses of each other:

$$R_{\text{passive}}(\theta) = R_{\text{active}}(\theta)^{-1} = R_{\text{active}}(-\theta)$$

This is why the quaternion formulas are conjugates of each other — $q\mathbf{v}q^*$ vs $q^*\mathbf{v}q$.

Concrete Example

Say **v** points along $\hat{x}$, and you apply a +90° rotation around $\hat{z}$:

- **Active:** the vector now points along $+\hat{y}$. Frame unchanged.
- **Passive:** the vector still points the same direction in space, but the new frame's $\hat{x}$-axis has rotated to where $\hat{y}$ used to be — so the vector's components are now $(0, -1, 0)$ in the new frame... wait:

More carefully:

	Active (+90° around z)	Passive (+90° around z)
Physical vector	Rotates to $+\hat{y}$	Unchanged in space
Frame	Unchanged	$\hat{x}_{new}$ points where old $\hat{y}$ was
New components	$(0, +1, 0)$	$(0, -1, 0)$

The components flip sign because if the frame rotated +90°, the vector *appears* to have gone −90° relative to it.

Why This Causes Confusion

1. **The math looks identical** — a 3×3 matrix or quaternion doesn't tell you which interpretation to use. Convention is everything.
2. **Different fields have different defaults** — robotics and aerospace often use passive (frame transforms), while graphics and physics often use active (moving objects).
3. **Composition order flips** — applying rotation A then B actively is $B \cdot A \cdot \mathbf{v}$, but the equivalent passive sequence may reverse order depending on convention.

The bottom line: the distinction isn't in the math — it's in **what you're asking the math to mean**.

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